

5.2.4 CHALLENGES

5.2.4.1 Issues related to sample preparation/isolation of microbes for sequencing

Despite the success of using metagenomic sequence data to guide the isolation and cultivation of microbes, there are still challenges and limitations that need to be addressed for more widespread success. Anaerobic microbes may also pose technical challenges for some targeted isolation methods due to limitations with cell sorters. Determining suitable growth media and physicochemical conditions for target organisms based solely on genome sequences can be challenging, as natural environment compositions are often unknown. Contamination can also be a significant challenge in microbiome research, as it can introduce non-target microbial DNA from external sources during sample collection, processing, and sequencing. Contamination can arise from reagents, equipment, and environmental sources, leading to false-positive or false-negative results. Proper controls, including negative controls and rigorous contamination prevention measures, should be implemented to minimize contamination and ensure accurate and reliable sequencing results.

Recovering genomic data from target microbes for cultivation strategies can also be impacted by issues such as incomplete cell extraction, cell lysis, or DNA degradation, which may affect the representation of certain species in metagenome data. DNA extraction is a critical step in sample preparation for microbiome sequencing, and different DNA extraction methods can yield different results. Some DNA extraction methods may be biased towards certain microbial groups or may not effectively lyse certain tough-to-break cells, resulting in the incomplete representation of the microbial community in the sequencing data.

5.3 TECHNOLOGY ANALYSIS

Sequencing by Synthesis (SBS)

Sequencing by synthesis (SBS) is a widely used technology in microbiome sequencing services for analyzing microbial communities. SBS is a high-throughput DNA sequencing method that enables the rapid and cost-effective sequencing of millions of DNA fragments in parallel. It has revolutionized the field of microbiome research by allowing researchers to explore the diversity and composition of microbial populations in each sample with high resolution.

The basic principle of SBS involves the use of DNA polymerases, enzymes that can synthesize DNA strands, to sequentially add nucleotides to a growing DNA chain. As each nucleotide is added, it is labeled with a fluorescent dye that corresponds to the base being incorporated. The fluorescence signal is then detected and recorded, allowing the identification of the nucleotide that was added. This process is repeated multiple times, typically in cycles, to generate DNA sequences that can be assembled into a complete microbial community profile.

Sequencing by Ligation (SBL)

Sequencing by ligation (SBL) is a DNA sequencing technology that is used in microbiome sequencing services for analyzing microbial communities. SBL is a high-throughput DNA sequencing method that relies on the ligation of short DNA oligonucleotide adapters to target DNA fragments, followed by their sequential hybridization and ligation to complementary oligonucleotides, enabling the determination of the DNA sequence.

The basic principle of SBL involves the use of DNA adapters that are ligated to the DNA fragments to be sequenced. These adapters contain known sequences at their ends that serve as priming sites for the sequencing reaction. The DNA fragments, together with the adapters, are then subjected to multiple rounds of hybridization and ligation with a set of fluorescently labeled oligonucleotides that are complementary to the known adapter sequences. The ligation events are detected and recorded, allowing the determination of the DNA sequence.

Pyrosequencing

Pyrosequencing, also known as 454 sequencing, is a DNA sequencing technology that has been used in microbiome sequencing services for studying the composition and function of microbial communities. Pyrosequencing is a high-throughput NGS method that allows for the rapid and parallel sequencing of DNA fragments, providing deep insights into the genetic diversity of microbiomes.

The basic principle of pyrosequencing involves the detection of pyrophosphate (PPi) released during DNA synthesis. In this technology, DNA fragments are first amplified using PCR with primers that contain unique barcodes or indexes to enable sample multiplexing. The amplified DNA fragments are then sequenced in a massively parallel manner using a DNA polymerase enzyme and a series of nucleotide substrates that are added sequentially. As each nucleotide is added, the enzyme incorporates it into the growing DNA chain, and if the added nucleotide is complementary to the template DNA, pyrophosphate (PPi) is released as a byproduct. The released PPi is then converted into light, which is detected and quantified, allowing for the determination of the DNA sequence.

Sanger Sequencing

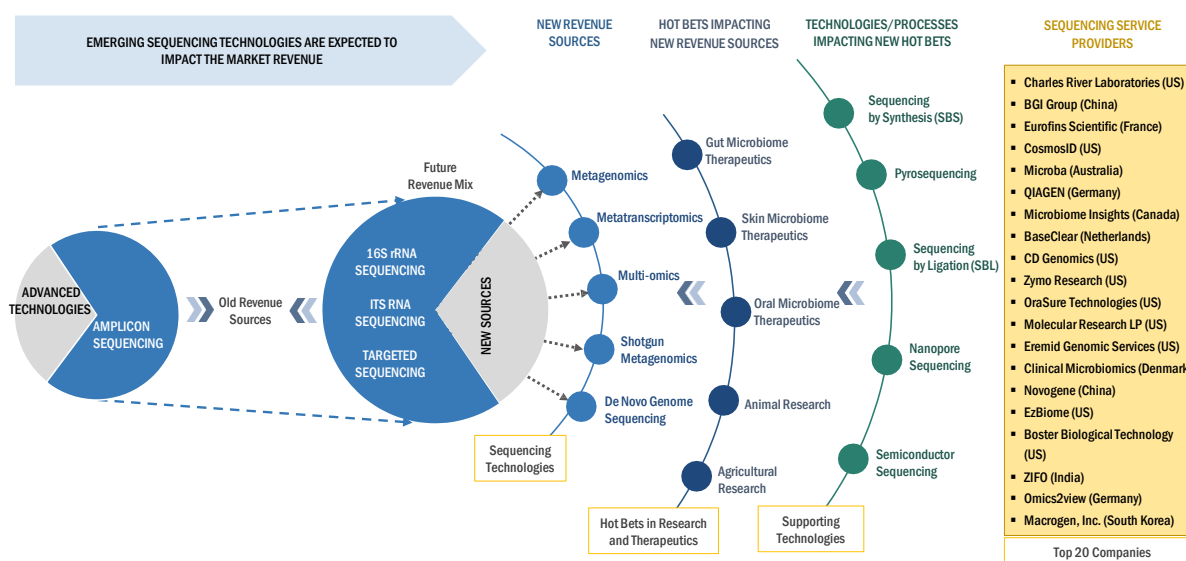
Sanger sequencing, also known as chain termination sequencing, is a traditional DNA sequencing technology used in microbiome sequencing services. While newer NGS technologies have gained popularity in recent years, Sanger sequencing still has its unique advantages and continues to be used in some microbiome research applications.

The basic principle of Sanger sequencing involves the enzymatic synthesis of DNA strands, where dideoxynucleotides (ddNTPs) are incorporated alongside regular deoxynucleotides (dNTPs) during DNA chain elongation. The incorporation of ddNTPs randomly terminates DNA synthesis, resulting in a set of DNA fragments of varying lengths that are complementary to the template DNA. The resulting DNA fragments are then separated by size using capillary electrophoresis, and the sequence is determined based on the order of the terminated ddNTPs.

5.4 TRENDS AND DISRUPTIONS IMPACTING CUSTOMERS' BUSINESSES

Robust research activities to explore the potential of the human microbiome present great potential in a number of treatments against obesity, inflammatory bowel disease, and diabetes, among other conditions. Information on microbiomes helps in the identification of the causal links between the microbiome and disease, further facilitating the development of efficient and targeted treatment strategies. Additionally, progress in state-of-the-art sequencing approaches, computational analyses, and experimental assays is supporting the growing demand for microbiome sequencing.

FIGURE 20 REVENUE SHIFT AND NEW REVENUE POCKETS FOR MICROBIOME SEQUENCING SERVICE PROVIDERS



Source: MarketsandMarkets analysis

The most commonly used sequencing technologies include 16S rRNA sequencing, ITS RNA sequencing, and whole-genome sequencing. Sequencing and experimental analyses of the microbiome continue to advance substantially. By coupling multi-omics technologies, statistical and computational analyses, and more advanced disease models, these approaches promisingly provide the opportunity to establish disease causality and subsequently inform therapeutic development.